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2024312506 余沅林抄 电磁场第3次作业

1.解: $\vec{D}_1 = \varepsilon_1 \vec{E}_1 = \varepsilon_0 \varepsilon_{r1}(20\vec{e}_x - 10\vec{e}_y + 50\vec{e}_z)$

$$\begin{cases} E_{1t} = E_{2t} \\ D_{1n} = D_{2n} \end{cases}$$

即 $\begin{cases} E_{2x} = E_{1x} = 20 \\ E_{2y} = E_{1y} = -10 \\ \varepsilon_2 E_{2z} = \varepsilon_1 E_{1z} \end{cases}$ 解得: $E_{2z} = \frac{\varepsilon_1}{\varepsilon_2} E_{1z} = \frac{\varepsilon_{r1}}{\varepsilon_{r2}} \cdot 50$

因此区域2的电场强度和电位移矢量为:

$$\vec{E}_2 = 20\vec{e}_x - 10\vec{e}_y + 50\frac{\varepsilon_{r1}}{\varepsilon_{r2}} \cdot \vec{e}_z$$

$$\vec{D}_2 = \varepsilon_2 \vec{E}_2 = \varepsilon_0 \varepsilon_{r2}(20\vec{e}_x - 10\vec{e}_y) + 50\varepsilon_0 \varepsilon_{r1} \cdot \vec{e}_z$$

2.解: 电场强度线的方向与电场强度矢量$\vec{E}$的方向一致, 因此在边界线上, 电场强度的法向分量为零, 即 $E_n = 0$

又因为 $E_n = -\frac{\partial \varphi}{\partial n}$, 所以该边界线是一条等位线, 其边界条件可表示为 $\frac{\partial \varphi}{\partial n} = 0$

3.解: 设空气的介电常数为 ε_0

(a). 设介质板内电场强度为 E, 空气区域电场强度为 E_0

$D = \varepsilon E = \varepsilon_0 E_0$ 极板间电压为 $U = E_0(d-t) + Et$

解得 $E = \frac{U\varepsilon_0}{\varepsilon(d-t) + \varepsilon_0 t}$ $E_0 = \frac{U\varepsilon}{\varepsilon(d-t) + \varepsilon_0 t}$

$$\sigma = D = \varepsilon E = \frac{U\varepsilon\varepsilon_0}{\varepsilon(d-t) + \varepsilon_0 t}$$

(b). $E = \frac{U}{d}$ $E_0 = \frac{U}{d}$ $\sigma = \varepsilon E = \frac{U\varepsilon}{d}$

$$\sigma_0 = \varepsilon_0 E_0 = \frac{U\varepsilon_0}{d}$$

4. 解: 设电位为 $\varphi(r)$, 则边值问题为

$$\begin{cases} \nabla^2 \varphi_1 = -\frac{\rho_1}{\varepsilon_0} & 0 < r < a \\ \nabla^2 \varphi_2 = -\frac{\rho_2}{\varepsilon_0} & a < r < b \\ \nabla^2 \varphi_3 = 0 & r > b \end{cases}$$

$$\begin{cases} \varphi_1(0) \text{ 有限} \\ \varphi_1(a) = \varphi_2(a),\ \varepsilon_0 \frac{\partial \varphi_1}{\partial r}\Big|_{r=a} = \varepsilon_0 \frac{\partial \varphi_2}{\partial r}\Big|_{r=a} \\ \varphi_2(b) = \varphi_3(b),\ \varepsilon_0 \frac{\partial \varphi_2}{\partial r}\Big|_{r=b} = \varepsilon_0 \frac{\partial \varphi_3}{\partial r}\Big|_{r=b} \\ \varphi_3(\infty) = 0 \end{cases}$$

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(2) 设电位为 $\varphi(r,\theta)$，则边值问题为

$$\begin{cases} \nabla^2\varphi_1=0 & a<r<b,\ 0<\theta<\frac{\pi}{2} \\ \nabla^2\varphi_2=0, & a<r<b,\ \frac{\pi}{2}<\theta<\pi \\ \varphi(b,\theta)=0 & \\ \varepsilon_1\frac{\partial\varphi_1}{\partial r}\big|_{r=a}+\varepsilon_2\frac{\partial\varphi_2}{\partial r}\big|_{r=a}=\frac{Q}{2\pi a^2} & \\ \varphi_1(a,\theta)=\varphi_2(a,\theta) & \theta=\pi/2 \end{cases}$$

(3). 设电位为 $\varphi(r)$，则边值问题为

$$\begin{cases} \nabla^2\varphi=0, & a<r<b \\ \varphi(b)=0 & \\ -\varepsilon\frac{\partial\varphi}{\partial r}\big|_{r=a}=\frac{\tau}{2\pi a} & \end{cases}$$

5. 解：设导体对地电容为 C_2，导体间电容为 C_1

(1) $\begin{cases} 3C_1=0.051\,\mu F & C_1=0.017\,\mu F \\ C_1+2C_2=0.037\,\mu F & C_2=0.01\,\mu F \end{cases}$ 导体1

(2) $C=C_1+3C_2=0.047\,\mu F$ (3) $Q=(C_1+C_2)\cdot U=2.7\,\mu C$

导体2的电荷为 $-2.7\,\mu C$

6. 解(1) 介质1和介质2均为 ε_1 镜像电荷

$$q_1'=\frac{\varepsilon_1-\varepsilon_2}{\varepsilon_1+\varepsilon_2}q_1$$

(2) 介质1和介质2均为 ε_1 镜像电荷

$$q_2''=\frac{2\varepsilon_1}{\varepsilon_1+\varepsilon_2}q_2$$

7. 解：内导体带单位长度电荷为 τ 外导体带 $-\tau$ 由高斯定理

介质内电场 $\vec{E}=\frac{\tau}{2\pi\varepsilon_0 r}\vec{e}_r$，$a<r<b$

内外导体间电压 $U=\int_a^b\vec{E}\cdot d\vec{r}=\frac{\tau}{2\pi\varepsilon_0}\ln\frac{b}{a}$

单位长度电容：$C=\frac{\tau}{U}=\frac{2\pi\varepsilon_0}{\ln(b/a)}$

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